1) The carrier signal $c(t) = 30Cos(\pi 10^4 t)$ is used in DSB-Large Carrier (LC)-AM.

a)(50 p.) If the message signal is $m(t) = 10 + 20Cos(2000\pi t) - 10Cos(4000\pi t)$ and efficiency is 11 %,

a1) find the modulation index, power of carrier and power of sidebands,

a2) find spectrum of the modulated signal $y(t)$ and sketch it, find its bandwidth,

a3) sketch noncoherent demodulator and state what condition RC must satisfy,

a4) sketch coherent demodulator and find the signals at each step in both time and frequency domains, sketch their spectra. Comment if DSB-LC-AM signal can be demodulated by coherent demodulator or not.

a1) $m_n(t) = \frac{m(t)}{mp}$ $\quad mp = 10+20+10 = 40$

$$m_n(t) = \frac{10}{40} + \frac{20}{40}\cos(2\pi 1kt) - \frac{10}{40}\cos(2\pi 2kt)$$
$$= \frac{1}{4} + \frac{1}{2}\cos(2\pi 1kt) - \frac{1}{4}\cos(2\pi 2kt)$$

$$P_{mn}(t) = \left(\frac{1}{4}\right)^2 + \frac{\left(\frac{1}{2}\right)^2}{2} + \frac{\left(\frac{1}{4}\right)^2}{2}$$
$$= \frac{1}{16} + \frac{1}{8} + \frac{1}{32} = \frac{2+4+1}{32} = \frac{7}{32}$$

$$V = \frac{\mu^2 P_{mn}}{1+\mu^2 P_{mn}} = 0,11$$

$$0,11 + 0,11\mu^2 P_{mn} = \mu^2 P_{mn}$$

$$\frac{\mu^2 P_{mn}(1-0,11)}{(0,89)} = \frac{0,11}{0,89}$$

$$\mu^2 = \frac{0,11}{0,89 \cdot \left(\frac{7}{32}\right)} = 0,565$$

$\Rightarrow \mu = 0,75$

$$P_c = \frac{30^2}{2} = 450\,W$$

$$P_s = \frac{\mu^2 A_c^2}{2} P_{mn}$$

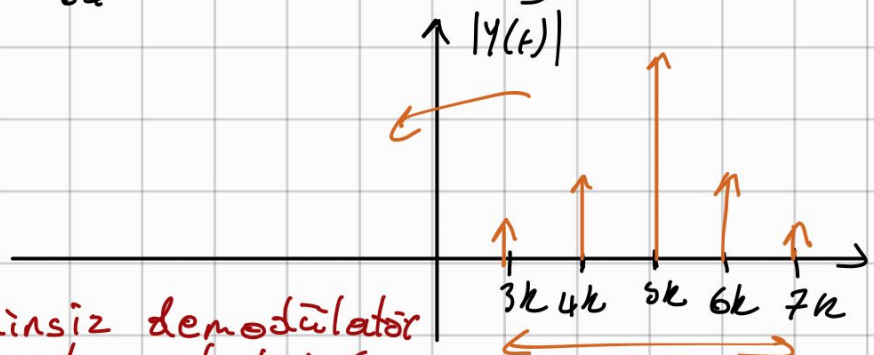
$$P_s = 55,62\,W$$

a2) $y(t) = A_c[1+\mu m_n(t)]\cos(2\pi f_c t)$ ($f_c = 5k$)

$$= 30\left[1 + \frac{3}{4}\left[\frac{1}{4} + \frac{1}{2}\cos(2\pi 1kt) - \frac{1}{4}\cos(2\pi 2kt)\right]\right]\cos(2\pi 5kt)$$

$$= 30\cos(2\pi 5kt) + \frac{3}{16}\cos(2\pi 5kt) + \frac{3}{8}\cos(2\pi 1kt)\cos(2\pi 5kt)$$
$$- \frac{3}{16}\cos(2\pi 2kt)\cos(2\pi 5kt)$$
$$= 30,19\cos(2\pi 5kt) + \frac{3}{16}\cos(2\pi 6kt) + \frac{3}{16}\cos(2\pi 4kt)$$
$$- \frac{3}{32}\cos(2\pi 7kt) - \frac{3}{32}\cos(2\pi 3kt)$$

$$Y(f) = \frac{30,19}{2}[\delta(f-5k)+\delta(f+5k)] + \frac{3}{32}[\delta(f-6k)+\delta(f+6k)]$$
$$+\frac{3}{32}[\delta(f-4k)+\delta(f+4k)] - \frac{3}{64}[\delta(f-7k)+\delta(f+7k)]$$
$$-\frac{3}{64}[\delta(f-3k)+\delta(f+3k)]$$



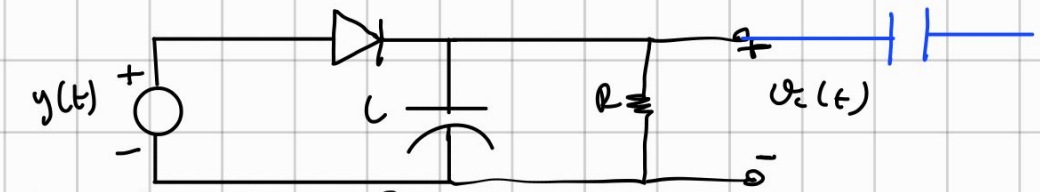
a3/

faz tahminsiz demodülatör
noncoherent modulator

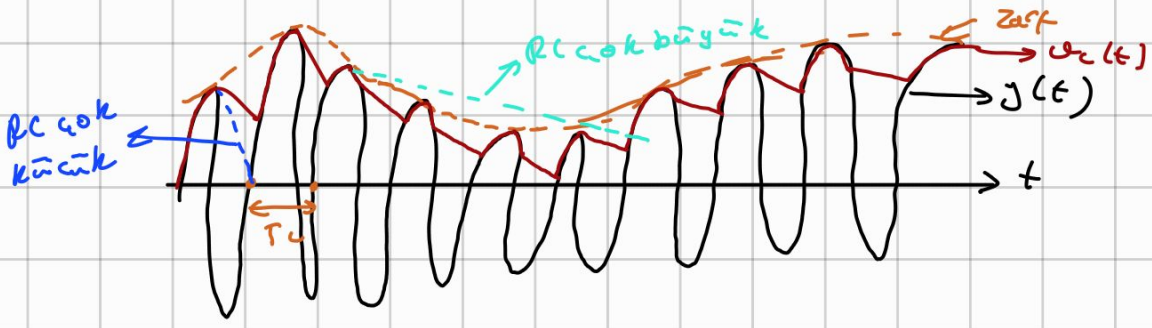
Zarf Detektörü

(Envelope detector)

$BG\{m(t)\} = B\,Hz$, $f_c >> B$



$y(t) = A_c[1 + k_a m(t)]\cos(2\pi f_c t)$



$$\frac{1}{f_c} << RC << \frac{1}{B}$$

$\frac{1}{f_c}$ ↓ $\frac{1}{5k}$ → $0,2 \cdot 10^{-3}$

$\frac{1}{B}$ ↓ $\frac{1}{4k}$ → $0,25 \cdot 10^{-3}$

mesajın zamandaki değişimimiz